

ENTERPRISE AI REFERENCE ARCHITECTURES

Transforming data centers with flexible, scalable,
and powerful NVIDIA RTX Pro 6000 and NVIDIA HGX
B300 GPU solutions



Redefining AI infrastructure with HPE and NVIDIA

Today's organizations are redefining what's possible with artificial intelligence (AI)—pushing boundaries in research, business, and innovation. At the center of this transformation stands the [NVIDIA AI Computing by HPE portfolio](#), a collaboration that unlocks new potential in enterprise AI infrastructure. By fusing robust HPE compute, software, and storage platforms with the latest advances in NVIDIA GPU accelerated computing hardware and software in jointly-validated configurations, blueprints make it easier to deploy reliable, scalable, and high-performance AI solutions.

This reference architecture (RA) overview offers two proven scalable solutions, from 16 to 16K GPUs. HPE tailors and optimizes solution components for distinct use cases, environments and operational scale. Inside HPE ProLiant servers, the first blueprint leverages the NVIDIA® RTX PRO 6000 Blackwell Server Edition and delivers an entry point into data center-scale AI with PCIe-based GPUs that balance performance, cost, and flexibility. Inside HPE XD servers and AI rack-scale systems, the second design showcases the power of the NVIDIA HGX B300 system—the high end of serious GPU computing. By presenting both options, this paper empowers organizations to select the ideal architecture for their unique needs, whether focusing more on AI modeling and training, AI inferencing, or deploying best-in-class GPU clusters from day one.

The collaboration between HPE and NVIDIA means each design is technically robust and backed by a collaborative ecosystem of support, management, and future-ready innovation. The result is a foundation that speeds time to insight for AI projects of any size, and enables enterprises to meet the evolving demands of AI, data analytics, and high-performance computing (HPC) workloads.

Reference architecture 1: Enterprise RA with NVIDIA RTX PRO 6000 Blackwell Server Edition

Ideal use cases: This cluster architecture is designed for organizations seeking a flexible, scalable entry point into AI workloads at data center scale. It is particularly well-suited for AI inference and fine tuning for models less than 70B parameters, and a range of enterprise applications where modular expansion and balanced cost-performance are essential. The design is optimized for environments prioritizing rapid deployment, reliability, and efficient scaling, making it a strong fit for enterprise data centers launching new AI initiatives or extending existing HPC capabilities.

The NVIDIA Enterprise Reference Architecture (Enterprise RA) using the RTX PRO 6000 Blackwell Server Edition GPUs is built to solve today's most challenging computational problems. This architecture employs a modular design, with each HPE ProLiant Compute DL380a equipped with up to eight RTX PRO 6000 Blackwell GPUs and NVIDIA Spectrum-X™ Networking to power next-generation AI and HPC use cases.



Key components:

- HPE ProLiant servers, each with up to 8 RTX PRO 6000 Blackwell Server Edition GPUs
- NVIDIA Spectrum-4 networking platform
- NVIDIA Connect-X-7 SuperNICs and DPUs
- NVIDIA AI Enterprise software

Architecture design:

- Composed of scalable units (SUs), each with 4 HPE ProLiant servers, allowing for rapid and flexible deployment at multiple scales
- Scales to 32 certified systems, for a total of 256 RTX PRO 6000 Blackwell GPUs in a fully tested system (8 SUs)
- Flexible, rail-optimized network architecture and support for custom rack layouts

Management and support:

- NVIDIA AI Enterprise, NVIDIA CUDA™, and NVIDIA Magnum IO™
- Hardware support via HPE fulfillment, software support via NVIDIA AI Enterprise subscription

Primary use cases:

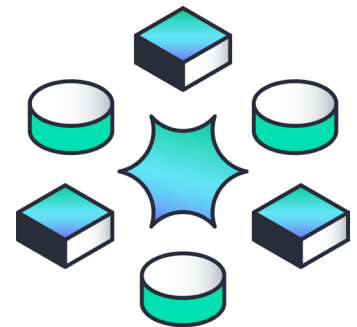
- AI inference and fine-tuning for models with up to 70B parameters, powering copilots, assistants and agentic AI
- Hybrid AI and single precision HPC workloads requiring modular, scalable infrastructure for high-performance data processing and simulation
- Data analytics, visual computing and engineering simulation to accelerate insight, design, and decision-making

This Enterprise RA provides a robust and adaptable framework for organizations to innovate confidently, supported by the combined strengths of HPE and NVIDIA engineering, software, and support ecosystem.

Reference architecture 2: Enterprise RA with NVIDIA HGX B300 and Spectrum-X Ethernet Networking

Ideal use cases: This cluster architecture is targeted at organizations requiring high levels of performance and scalability for mission-critical AI workloads. Optimized for multi-node AI training, large-scale inference, and hybrid applications, HPE XD servers with NVIDIA HGX B300 GPUs are the choice for enterprises ready to deploy advanced GPU clusters capable of handling demanding models and large data sets. With its massive scaling potential, this blueprint supports everything from scientific research and engineering simulations to global business analytics and next-generation AI development.

The NVIDIA Enterprise RA leverages HPE XD servers with NVIDIA HGX B300 GPUs and Spectrum-X Ethernet platform to drive the next generation of data center infrastructure for AI and HPC. This architecture is built to solve the most challenging computational problems facing enterprises today, offering certified configurations that can be deployed globally.



Key components:

- HPE Compute XD690 servers, each with 8 NVIDIA HGX B300 SXM GPUs
- NVIDIA Spectrum-4 networking platform
- NVIDIA ConnectX-8 SuperNICs and BlueField-3 DPUs
- NVIDIA AI Enterprise software

Architecture design:

- Modular building blocks: SUs, each containing four HPE XD servers for rapid deployment and flexible scaling
- Scales up to 128 certified HGX B300 systems, totaling 1024 HGX B300 SXM GPUs (32 SUs fully tested); larger clusters are possible to meet customer requirements
- Rail-optimized end-of-row network architecture, adaptable to varying rack layouts and HPE XD server counts

Management and support:

- NVIDIA AI Enterprise, NVIDIA CUDA, and NVIDIA Magnum IO™
- Hardware support via HPE fulfillment, software support via NVIDIA AI Enterprise subscription

Primary use cases:

- AI inference—Large per-node and medium per-GPU model parameter inference workloads
- AI training—Large to small model training and fine tuning based on cluster sizing
- Hybrid, multi-node AI and HPC deployments demanding maximum scalability and performance

This RA delivers a modular, scalable, and high-performance solution for organizations seeking to deploy advanced AI infrastructure with confidence. Backed by the combined expertise and support systems of HPE and NVIDIA, it empowers enterprises to push the boundaries of AI, HPC, and data analytics.

Blueprints with NVIDIA: A platform for accelerated innovation

Reference architectures with NVIDIA represents a new era of enterprise-ready AI—designs to streamline AI adoption, accelerate deployment, and scale innovation across a wide range of industries and use cases. The resulting HPE AI solutions with NVIDIA provide a holistic environment where compute, software, storage, networking, and server GPU acceleration can be integrated to create an agile AI infrastructure.

Central to these architectures is a philosophy of modularity and scalability. Organizations can start with a single scalable unit and expand to hundreds of nodes as needs evolve, thanks to a flexible design that adapts to both current and future demands. The platform integrates high-performance HPE ProLiant and XD servers with NVIDIA GPUs, from the versatile RTX PRO 6000 Blackwell GPU, ideal for mainstream AI workloads and rapid experimentation, to the groundbreaking HGX B300, purpose-built for advanced AI and HPC.

Automation and orchestration are core to the HPE experience. With management solutions such as NVIDIA AI Enterprise, CUDA and Magnum IO, so workloads are efficiently deployed, monitored, and optimized. Enterprises gain access to a comprehensive suite of tools for model training, inference, and analytics, all supported by robust hardware and software maintenance from HPE and NVIDIA partners.

The portfolio of available designs integrates curated software with secure, validated infrastructure to provide an engineering-validated stack designed for today's AI workloads.

Collaboration is another hallmark of these enterprise blueprints. The collaboration between HPE and NVIDIA brings together expertise in solution engineering. The installation and deployment expertise from HPE gives enterprise clients the confidence to deploy AI at scale. The designs' modular nature not only simplifies expansion, but also empowers organizations to integrate their AI workloads with broader IT initiatives.

For industries ranging from healthcare and life sciences to finance, manufacturing, and retail, HPE enterprise AI solutions with NVIDIA enable faster time to insight, greater operational efficiency, and the freedom to innovate. With engineering-validated designs, every enterprise, regardless of size or specialization, can harness the transformative power of AI.

Ultimately, HPE enterprise AI architectures with NVIDIA stand as a foundation for next-generation AI, designed for those determined to lead in an increasingly data-driven world. It is the key to unlocking new possibilities, automating complex processes, and driving breakthroughs across business and research.





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Why HPE for AI solutions

Many teams face familiar barriers, such as fragmented data, security concerns, and infrastructure that can't keep pace, making it difficult to move from AI pilot to production with confidence. HPE delivers AI solutions built for the realities of the enterprise, helping organizations progress from ambition to outcomes even when budgets, skills, and operational capacity are constrained. The HPE approach focuses on creating a strong, unified AI foundation designed for secure, sustainable, enterprise-scale deployment across hybrid environments.

HPE leads with experience and expertise to guide the workload-focused decision points that drive a right-sized design—what must run now versus later, which workloads require acceleration, and how to operationalize AI without adding complexity. With an extensive portfolio and services that emphasize trusted data pipelines, AI-tuned operations, and turnkey hybrid platforms, HPE helps speed time to AI value while preserving flexibility and control, enabling organizations to turn AI initiatives into repeatable, measurable business outcomes.

A foundation for the future of AI

HPE enterprise AI solutions with NVIDIA set a new standard for enterprise-ready, flexible, and future-proof AI deployment. With these reference architectures—one built for rapid entry into data center AI using HPE ProLiant servers with PCIe-based RTX PRO 6000 Blackwell GPUs, the other harnessing the unmatched performance of HPE XD servers and AI rack-scale systems with NVIDIA HGX B300 GPUs—organizations have the freedom to innovate without compromise.

Whether pursuing AI inferencing, fine-tuning, large-scale AI training, or deployment of diverse workloads, the combination of engineering expertise from HPE and NVIDIA enables seamless scaling, simplified management, and sustained performance. As AI ambitions grow, these designs serve as a launching pad—backed by best-in-class support and a shared vision for what AI can achieve in the enterprise.

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